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(Liquid + liquid) extraction of methanol from alkanes using dialkylphosphate-based ionic liquids as solvents

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ABSTRACT

In this work, the feasibility of ionic liquids (ILs), 1,3-dimethylimidazolium dimethylphosphate ([MMIM][DMP]), 1-ethyl-3-methylimidazolium diethylphosphate ([EMIM][DEP]), and 1-butyl-3-methylimidazolium dibutylphosphate ([BMIM][DBP]), as solvents for the extraction of methanol from its mixtures with hexane and heptane was analyzed. The knowledge of (liquid + liquid) equilibria (LLE) of these mixtures is necessary for the design of the extraction separation process. Hence, the LLE data for the ternary systems, {methanol + hexane + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP])}, and {methanol + heptane + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP])}, were measured at $T = 298.2$ K and atmospheric pressure. The experimental results were correlated with the thermodynamic nonrandom two-liquid (NRTL) model. The solute distribution ratios of methanol and methanol/alkane selectivities, derived from the experimental LLE data, were calculated and analyzed to evaluate the capability of the studied ILs to accomplish the separation target. Meanwhile, these capabilities were also compared with that of other ILs obtained from the literature.

1. Introduction

Alcohols and alkanes mixtures are present in industrial production processes of oxygenated additives for unleaded gasoline [1]. Separation of these mixtures is very challenging because of the proximity of their boiling points as well as the formation of azeotropes. The extractive distillation is one of the most widely used techniques to separate these mixtures [2]; however, it is energetically unfavorable. (Liquid + liquid) extraction is a beneficial separation process because it can be carried out under ambient conditions and the energy consumption and environmental impact can be greatly reduced. But the selection of efficient and suitable solvents with high selectivity is paramount to ensure an economic operation.

In the past decades, ionic liquids (ILs) have received considerable attention for their use as potential substitutes for classical organic solvents in the extraction processes [3–7]. Their unique properties include negligible vapor pressure at room temperature, wide liquid range, high thermal/chemical stability, excellent solubility for organic/inorganic compounds, and superior regeneration ability [8,9]. The replacement of volatile organic solvents by nonvolatile ILs can offer several advantages such as less complex process or simpler regeneration of the solvent.

Currently, there are a few publications concerning the ILs as solvents for the separation of alcohols and alkanes. Pereiro *et al.* [10,11] studied the separation of ethanol from hexane and heptane using the alkylsulfate-based ILs as solvents. Revelli *et al.* [12] published the LLE data for the ternary systems containing imidazolium-based ILs with (*n*-alcohols + heptane) at $T = 298.15$ K and atmospheric pressure. González and co-workers [13–15] focused on the effect of the alkyl substituent length of imide and dicyanamide-based ILs, in the separation of alcohols and alkanes by solvent extraction. Marciniak and co-workers [16,17] analyzed the effect of fluorine-containing ILs on the LLE data of (methanol + heptane).

As a continuation of our work about the application of ILs in the separation processes [18–23], three dialkylphosphate-based ILs, [MMIM][DMP], [EMIM][DEP], and [BMIM][DBP], were used as solvents for the (liquid + liquid) extraction of methanol from its mixtures with hexane and heptane. In order to understand the role of ILs in the separation process, LLE data are required. Therefore, in this work, the experimental LLE data were measured for the ternary systems, {methanol + hexane + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP])}, and {methanol + heptane + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP])}, at $T = 298.2$ K and atmospheric pressure. The NRTL [24] model was used to correlate the

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experimental results for the studied ternary systems. The capability of the studied ILs as solvents in the (liquid + liquid) extraction was analyzed by using the solute distribution ratios of methanol and methanol/alkane selectivities. Meanwhile, these capabilities were also compared with that of other ILs obtained from the literature.

2. Experimental

2.1. Chemicals and materials

Methanol (Sinopharm Chemical Reagent Co., Ltd., Shanghai, China), hexane (Sinopharm Chemical Reagent Co., Ltd., Shanghai, China), heptane (Aladdin Industrial Corporation, Shanghai, China), and 1-propanol (Aladdin Industrial Corporation, Shanghai, China) were used in this work with stated purities of 99.5% (mass), 99.0% (mass), 99.0% (mass), and 99.9% (mass), respectively. The studied ILs, [MMIM][DMP], [EMIM][DEP] and [BMIM][DBP], were synthesized in the laboratory according to the procedure in the literature [25] as described in our previous work [20]. These ILs were dried by heating at $T = 393.2$ K under high vacuum (0.2 kPa) for 48 h prior to use. The mass fraction of water in the IL, measured by Karl Fischer titration (AKF-2010, China), was $x_w < 0.0005$. The structures of synthesized ILs were verified by ^{1}H NMR spectrum (Bruker AV-600 NMR spectrometer) with certified purity higher than 98.0% mass fraction. The specifications of the chemicals used in this work are given in table 1. The structure formulas of the studied ILs are shown in figure 1. All chemicals are kept in a glove box under an inert nitrogen atmosphere to prevent moisture.

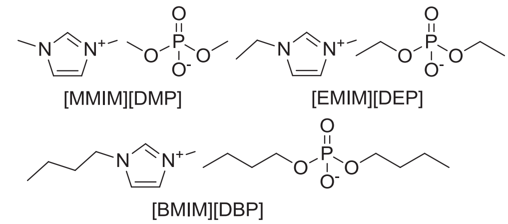

FIGURE 1. Structure formulas of the ionic liquids studied in this work.

2.2. Apparatus and experimental procedure

The determination of experimental LLE data was performed in a 50 mL glass cell containing a magnetic stirrer and thermostatically controlled at $T = 298.2$ K. A schematic diagram of the experimental apparatus is shown in figure 2. For the experimental measurements of LLE data, 30 mL of an immiscible ternary mixture of known composition was put into the glass cell and stirred using the magnetic stirrer. The temperature in the glass cell was measured with a precise and calibrated thermometer with a standard uncertainty of $T = 0.1$ K. The sample mixture was stirred vigorously for 1 h with a speed of 800 rpm (time dependency experiments showed that equilibrium for the system is reached within 1 h) to intimate contact between both phases. Then, the final two-phase solution was left to settle for 4 h to ensure a complete split between the equilibrium phases.

Then the samples were obtained from the upper and lower layers using a syringe, and compositional analysis was carried out. The compositions of methanol and alkanes (*i.e.*, hexane and heptane) in the sample were analyzed by gas chromatography. The gas chromatograph (GC-6890, China) was equipped with a flame ionization detector and the capillary column was SE-54 (50 m × 0.32 mm). The carrier gas was high purity nitrogen (Nanjing special gas factory Co., Ltd., Nanjing, China, ⩾99.999%), and the operating conditions were as follows: the injector temperature was 423.2 K, the oven temperature was 343.2 K, and the detector temperature was 473.2 K. 1-Propanol was added to the sample as an internal standard substance for the GC analysis, and a calibration curve was obtained from a set of gravimetrically prepared standard solutions, which allowed for the quantification of methanol and alkanes in the samples. Measurements were carried out in duplicate to exclude exceptions in the composition analysis. Each sample was analyzed five times. An analytical balance with a standard uncertainty of 0.0001 g was used to weigh the sample. Ternary mixtures with a well-know composition were prepared by mass and were analyzed with the GC. The compositions from GC analysis were compared with those obtained by mass. The standard uncertainties of the mole fractions of methanol and alkanes in the samples were 0.001.

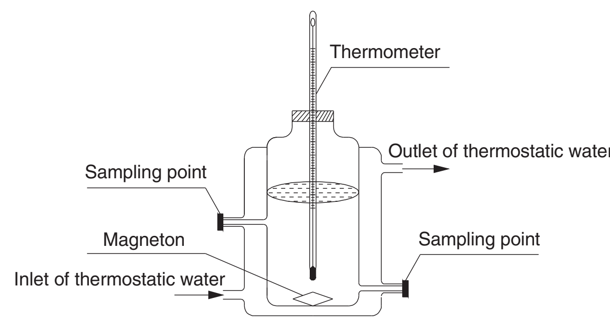

FIGURE 2. Schematic diagram of the experimental apparatus.

Because the ILs have negligible vapor pressure, they cannot be analyzed by GC. The whole of the ILs were retained by an empty precolumn located between the injector and the capillary column

TABLE 1
Sources, purities, purification methods, and water contents by mass, w_w, of the chemicals used in this work.

Chemical	Source	Mass fraction purity	Purification method	$w_w/10^{-6}$	Analysis method
Methanol	Sinopharm Chemical Reagent Co., Ltd.	⩾0.995	None		
Hexane	Sinopharm Chemical Reagent Co., Ltd.	⩾0.990	None		GC[d]
Heptane	Aladdin Industrial Corporation	⩾0.990	None		GC[d]
1-Propanol	Aladdin Industrial Corporation	⩾0.999	None		GC[d]
[MMIM][DMP][a]	Prepared in the laboratory	⩾0.980	Vacuum desiccation	<500	KF[e], NMR[f]
[EMIM][DEP][b]	Prepared in the laboratory	⩾0.980	Vacuum desiccation	<500	KF[e], NMR[f]
[BMIM][DBP][c]	Prepared in the laboratory	⩾0.980	Vacuum desiccation	<500	KF[e], NMR[f]

[a] [MMIM][DMP] = 1,3-dimethylimidazolium dimethylphosphate.
[b] [EMIM][DEP] = 1-ethyl-3-methylimidazolium diethylphosphate.
[c] [BMIM][DBP] = 1-butyl-3-methylimidazolium dibutylphosphate.
[d] Gas chromatography.
[e] Karl Fischer titration.
[f] Nuclear magnetic resonance.

during the GC analysis. The empty precolumn was periodically cleaned to prevent the ILs from coming into the capillary column. The ^{1}H NMR spectrum analysis was used to check that the IL was absent in the upper layer (*i.e.*, alkane-rich phase) so that the IL in this phase could be assumed as zero. This assumption has also been shown by other authors studying the LLE data of the systems (methanol + alkane) with different ILs [15–17]. The composition of IL in the lower layer (*i.e.*, IL-rich phase) was measured gravimetrically by determining the mass difference of the liquid sample (≈3.5 g) before and after vaporization of the solvents at T = 393.2 K and under high vacuum (0.2 kPa) until constant mass [26–28]. The purity of the IL after vaporization was verified by ^{1}H NMR spectrum. Meanwhile, no presence of solvents in the IL was observed with liquid chromatography (LC-20AT, SHIMADZU). The standard uncertainty of the mole fraction of IL in the samples was found to be 0.002.

3. Results and discussion

3.1. Experimental LLE data

The experimental LLE data of the ternary systems {methanol (1) + hexane (2) + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP]) (3)}, and {methanol (1) + heptane (2) + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP]) (3)} at T = 298.2 K and atmospheric pressure are respectively tabulated in tables 2 and 3, where x_i represents the mole fraction of component i in the liquid phase. The triangular diagrams with the experimental tie-lines of the studied ternary systems are also plotted in figures 3 and 4, where the differences in the size of the immiscibility region and the slopes of the experimental tie-lines are displayed. As shown in triangular diagrams, the immiscibility region decreases as the size of the alkyl chain length of the investigated ILs increases, showing the same behavior in the literature [13–15]. The amount of the alkanes (*i.e.*, hexane and heptane) dissolved in the ILs is small, and the solubility of the alkanes in the studied ILs decreases in the following order: [BMIM][DBP] > [EMIM][DEP] > [MMIM][DMP]. Besides, heptane shows a lower solubility than hexane in the studied ILs. Finally, the positive slopes of the tie-lines on the triangular diagrams indicate that methanol has a higher affinity toward the studied ILs than toward the hexane or heptane.

TABLE 2
Experimental liquid–liquid equilibria data on mole fraction, x, the solute distribution ratios of methanol, β, and methanol/hexane selectivities, S, for the studied ternary systems at T = 298.2 K and p = 101.3 kPa.[a]

Upper layer		Lower layer		β	S
x_1^{I}	x_2^{I}	x_1^{II}	x_2^{II}		
Methanol (1) + Hexane (2) + [MMIM][DMP] (3)					
0.000	1.000	0.000	0.040		
0.002	0.998	0.129	0.041	64.5	1570.0
0.003	0.997	0.199	0.045	66.3	1469.7
0.005	0.995	0.288	0.046	57.6	1245.9
0.008	0.992	0.384	0.047	48.0	1013.1
0.014	0.986	0.553	0.049	39.5	794.8
0.019	0.981	0.678	0.051	35.7	686.4
0.023	0.977	0.751	0.053	32.7	601.9
0.030	0.970	0.824	0.054	27.5	493.4
Methanol (1) + Hexane (2) + [EMIM][DEP] (3)					
0.000	1.000	0.000	0.061		
0.003	0.997	0.116	0.062	38.7	621.8
0.006	0.994	0.237	0.067	39.5	586.0
0.010	0.990	0.348	0.068	34.8	506.6
0.014	0.986	0.455	0.070	32.5	457.8
0.022	0.978	0.551	0.073	25.0	335.5
0.029	0.971	0.643	0.075	22.2	287.1
0.033	0.967	0.698	0.077	21.2	265.6
0.042	0.958	0.781	0.080	18.6	222.7
Methanol (1) + Hexane (2) + [BMIM][DBP] (3)					
0.000	1.000	0.000	0.083		
0.003	0.997	0.104	0.085	34.7	406.6
0.007	0.993	0.248	0.099	35.4	355.4
0.010	0.990	0.316	0.103	31.6	303.7
0.015	0.985	0.440	0.109	29.3	265.1
0.025	0.975	0.551	0.117	22.0	183.7
0.033	0.967	0.629	0.125	19.1	147.5
0.041	0.959	0.694	0.132	16.9	123.0
0.050	0.950	0.747	0.143	14.9	99.3

[a] Standard uncertainties u are $u(T)$ = 0.1 K, $u(p)$ = 1 kPa, $u(x_1)$ = 0.001 and $u(x_2)$ = 0.001.

TABLE 3
Experimental liquid–liquid equilibria data on mole fraction, x, the solute distribution ratios of methanol, β, and methanol/heptane selectivities, S, for the studied ternary systems at T = 298.2 K and p = 101.3 kPa.[a]

Upper layer		Lower layer		β	S
x_1^{I}	x_2^{I}	x_1^{II}	x_2^{II}		
Methanol (1) + Heptane (2) + [MMIM][DMP] (3)					
0.000	1.000	0.000	0.006		
0.002	0.998	0.108	0.009	54.0	5988.0
0.003	0.997	0.175	0.011	58.3	5287.1
0.005	0.995	0.257	0.012	51.4	4261.9
0.008	0.992	0.369	0.014	46.1	3268.3
0.011	0.989	0.455	0.015	41.4	2727.2
0.016	0.984	0.584	0.016	36.5	2244.8
0.022	0.978	0.692	0.017	31.5	1809.6
0.027	0.973	0.815	0.019	30.2	1545.8
Methanol (1) + Heptane (2) + [EMIM][DEP] (3)					
0.000	1.000	0.000	0.009		
0.003	0.997	0.094	0.011	31.3	2839.9
0.005	0.995	0.183	0.015	36.6	2427.8
0.008	0.992	0.269	0.018	33.6	1853.1
0.011	0.989	0.342	0.021	31.1	1464.2
0.017	0.983	0.466	0.023	27.4	1171.6
0.024	0.976	0.606	0.027	25.3	912.7
0.029	0.971	0.704	0.031	24.3	760.4
0.035	0.965	0.792	0.036	22.6	606.6
Methanol (1) + Heptane (2) + [BMIM][DBP] (3)					
0.000	1.000	0.000	0.025		
0.003	0.997	0.089	0.033	29.7	896.3
0.006	0.994	0.199	0.043	33.2	766.7
0.010	0.990	0.288	0.050	28.8	570.2
0.014	0.986	0.377	0.059	26.9	450.0
0.022	0.978	0.517	0.066	23.5	348.2
0.029	0.971	0.636	0.075	21.9	283.9
0.037	0.963	0.718	0.084	19.4	222.5
0.046	0.954	0.756	0.098	16.4	160.0

[a] Standard uncertainties u are $u(T)$ = 0.1 K, $u(p)$ = 1 kPa, $u(x_1)$ = 0.001 and $u(x_2)$ = 0.001.

3.2. Correlation of LLE data

The NRTL model was used to correlate the experimental LLE data for the studied ternary systems. This model has been proved a good capacity in correlating the LLE data of ternary systems containing ILs [29–32]. The activity coefficients in multicomponent systems were calculated by the NRTL model:

$$\ln \gamma_i = \frac{\sum_{j=1}^{m} \tau_{ji} G_{ji} x_j}{\sum_{l=1}^{m} G_{li} x_l} + \sum_{j=1}^{m} \frac{x_j G_{ij}}{\sum_{l=1}^{m} G_{lj} x_l} \left(\tau_{ij} - \frac{\sum_{r=1}^{m} x_r \tau_{rj} G_{rj}}{\sum_{l=1}^{m} G_{lj} x_l} \right), \tag{1}$$

where

$$G_{ij} = \exp(-\alpha_{ij} \tau_{ij}), \tag{2}$$

$$\tau_{ij} = \frac{\Delta g_{ij}}{RT}, \tag{3}$$

where x_i is the mole fraction of component i in the liquid phase; α_{ij} is nonrandomness factor; Δg_{ij} is binary interaction parameter; and T is temperature.

For the studied ternary systems, the parameters of the NRTL model (*i.e.*, Δg_{ij}, Δg_{ji}, and α_{ij}) were obtained by minimization of the objective function (OF) given in equation (4), using the SOLVER tool of the Microsoft Excel 2003 spreadsheet [33].

$$OF = \sum \left[\left(x_1^{\mathrm{I}} - x_1^{\mathrm{I}}(calc)\right)^2 + \left(x_2^{\mathrm{I}} - x_2^{\mathrm{I}}(calc)\right)^2 + \left(x_1^{\mathrm{II}} - x_1^{\mathrm{II}}(calc)\right)^2 + \left(x_2^{\mathrm{II}} - x_2^{\mathrm{II}}(calc)\right)^2 \right], \qquad (4)$$

where x_1^{I}, x_2^{I}, x_1^{II}, x_2^{II} and $x_1^{\mathrm{I}}(calc)$, $x_2^{\mathrm{I}}(calc)$, $x_1^{\mathrm{II}}(calc)$, $x_2^{\mathrm{II}}(calc)$ are the experimental and calculated mole fractions, respectively; superscripts I and II represent the upper and lower layers, respectively; subscripts 1 and 2 represent the components methanol and alkane, respectively; and the summations are extended to the whole range of data points.

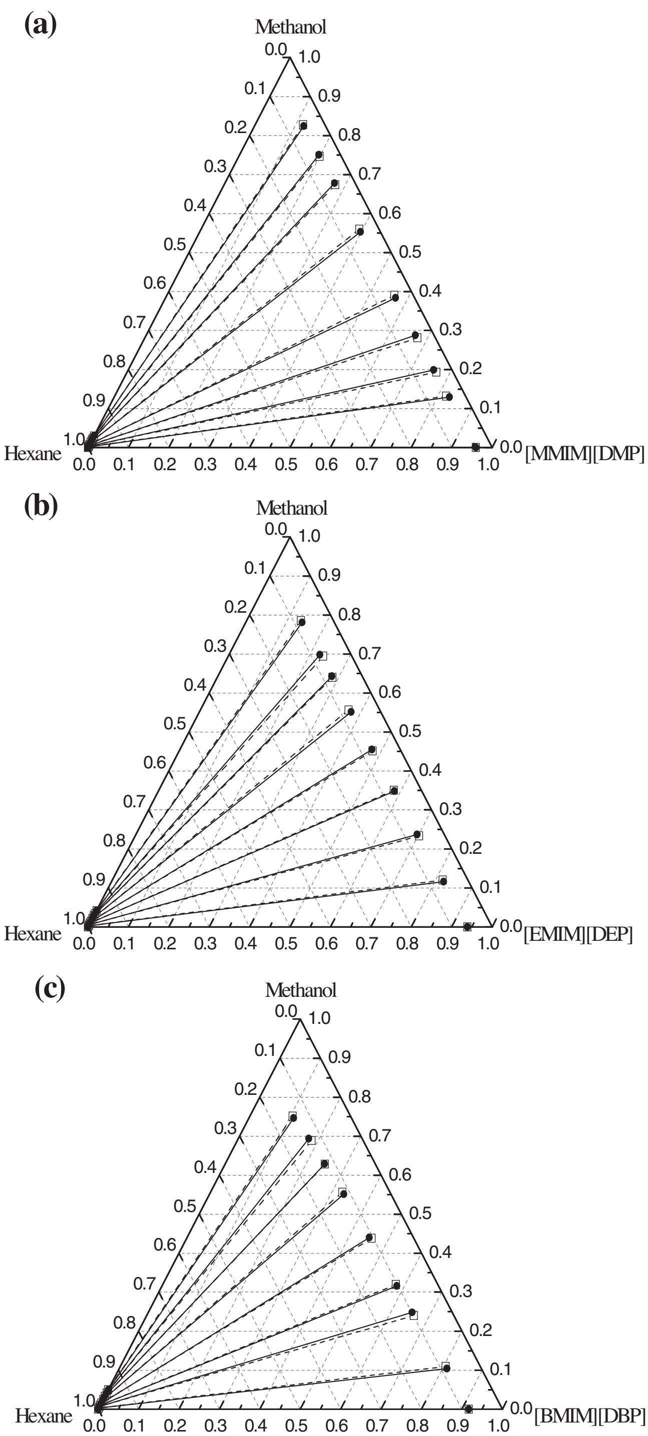

FIGURE 3. Experimental and calculated LLE data in mole fraction for the studied ternary systems at T = 298.2 K and atmospheric pressure: (a) methanol (1) + hexane (2) + [MMIM][DMP] (3); (b) methanol (1) + hexane (2) + [EMIM][DEP] (3); (c) methanol (1) + hexane (2) + [BMIM][DBP] (3). Solid lines and full points represent experimental tie-lines, and dashed lines and empty squares represent calculated data by the NRTL model.

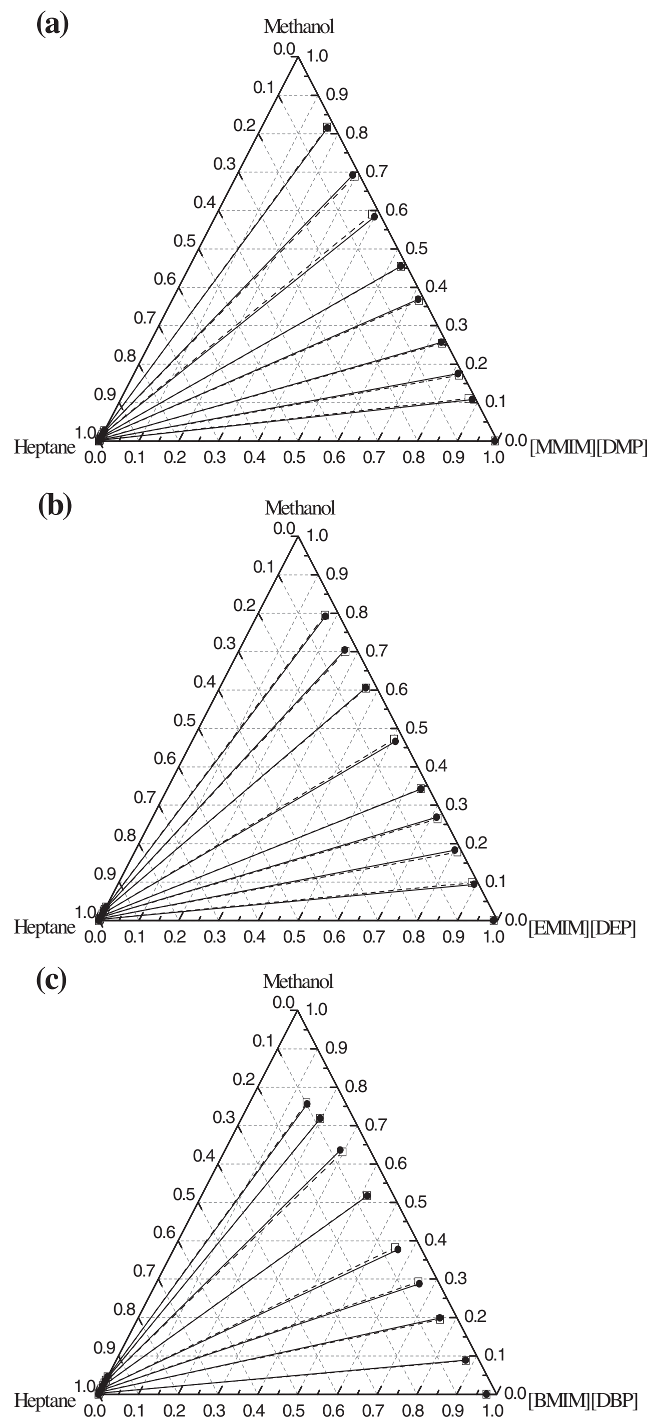

FIGURE 4. Experimental and calculated LLE data in mole fraction for the studied ternary systems at T = 298.2 K and atmospheric pressure: (a) methanol (1) + heptane (2) + [MMIM][DMP] (3); (b) methanol (1) + heptane (2) + [EMIM][DEP] (3); (c) methanol (1) + heptane (2) + [BMIM][DBP] (3). Solid lines and full points represent experimental tie-lines, and dashed lines and empty squares represent calculated data by the NRTL model.

TABLE 4
Binary interaction parameters, Δg_{ij} and Δg_{ji}, nonrandomness factor, α_{ij}, and root mean square deviations, σ, obtained from the correlation of the experimental (liquid + liquid) equilibria data of the studied ternary systems by the NRTL model at T = 298.2 K and p = 101.3 kPa.

i–j	Δg_{ij}/J · mol^{-1}	Δg_{ji}/J · mol^{-1}	α_{ij}	σ
	Methanol (1) + Hexane (2) + [MMIM][DMP] (3)			
1–2	10388.9	9261.6	0.445	0.307
1–3	22145.2	−15822.1	0.041	
2–3	10486.2	10195.1	0.447	
	Methanol (1) + Hexane (2) + [EMIM][DEP] (3)			
1–2	8454.3	10741.5	0.420	0.269
1–3	33608.8	942.9	0.199	
2–3	89625.6	10832.0	0.304	
	Methanol (1) + Hexane (2) + [BMIM][DBP] (3)			
1–2	6618.3	10085.5	0.409	0.277
1–3	51372.3	806.1	0.153	
2–3	69803.8	9318.5	0.247	
	Methanol (1) + Heptane (2) + [MMIM][DMP] (3)			
1–2	12130.8	15010.1	0.445	0.287
1–3	52263.1	−32688.9	0.015	
2–3	18904.7	14681.7	0.467	
	Methanol (1) + Heptane (2) + [EMIM][DEP] (3)			
1–2	10263.6	10283.7	0.452	0.274
1–3	8589.9	765.9	0.419	
2–3	21793.4	13463.9	0.429	
	Methanol (1) + Heptane (2) + [BMIM][DBP] (3)			
1–2	6743.5	12997.1	0.376	0.281
1–3	68891.4	2878.8	0.102	
2–3	28311.7	10558.8	0.337	

The fitting parameters of the NRTL model and the values of the root mean square deviations of liquid phase mole fraction, σ, are presented in table 4. These root mean square deviations were calculated by applying the following equation:

$$\sigma = 100\left(\frac{\sum\left(x_{ilm}^{exptl} - x_{ilm}^{calc}\right)^2}{6k}\right)^{1/2}, \quad (5)$$

where x is the mole fraction; superscripts *exptl* and *calc* represent the experimental and calculated values, respectively; subscripts i, l and m represent the component, the phase and the tie-lines, respectively; k represents the number of experimental tie-lines; and the summations are extended to the whole range of data points. From the low values of σ listed in table 4, it can be concluded that the NRTL model can correlate the experimental LLE data with a good degree of accuracy. On the other hand, for a visual confirmation the experimental tie-lines and those calculated with the NRTL model are shown in figures 3 and 4, where the good of the correlations can also be confirmed.

3.3. Solute distribution ratio and selectivity

In order to evaluate the extraction capability of the studied ILs as solvents, solute distribution ratios of methanol and methanol/alkane selectivities are also given in tables 2 and 3. The solute distribution ratio provides the solvent capacity of the IL, which is related to the amount of IL required for the extraction process. The selectivity is an important parameter which is widely used to assess the feasibility of utilizing the IL in the (liquid + liquid) extraction. High values of these parameters are necessary to regard an IL a good candidate for an extraction process. The solute distribution ratios of methanol, β, and methanol/alkane selectivities, S, are defined as follows:

$$\beta = \frac{x_1^{II}}{x_1^{I}}, \quad (6)$$

$$S = \frac{x_1^{II}x_2^{I}}{x_1^{I}x_2^{II}}, \quad (7)$$

where x_1^I and x_2^I are respectively the mole fractions of methanol and alkane in the upper layer (*i.e.*, alkane-rich phase); and x_1^{II} and x_2^{II} are respectively the mole fractions of methanol and alkane in the lower layer (*i.e.*, IL-rich phase).

The values of β and S as a function of the methanol mole fraction in the upper layer are plotted in figures 5 and 6. As can be observed from figures 5 and 6, on the whole, the values of β and S for the studied ternary systems decrease as the quantity of methanol in the upper layer increase. The high values of β and S are obtained for the studied ILs, especially at low compositions of methanol in the upper layer. Comparing the ternary systems {methanol (1) + alkane (2) + IL (3)}, the values of β and S for the studied ILs decrease in the following order: [MMIM][DMP] > [EMIM][DEP] > [BMIM][DBP], which is in concordance with the increasing size of the immiscibility region. This order shows that the alkyl chain length plays a negative role in the capability of the studied ILs to removal methanol from its mixtures with hexane and heptane. This behavior was also observed in the literature [15]. Moreover, comparing the ternary systems studied with the same IL and different alkanes, it is possible to observe that the ILs in the systems with heptane show higher values of S than the ILs in the

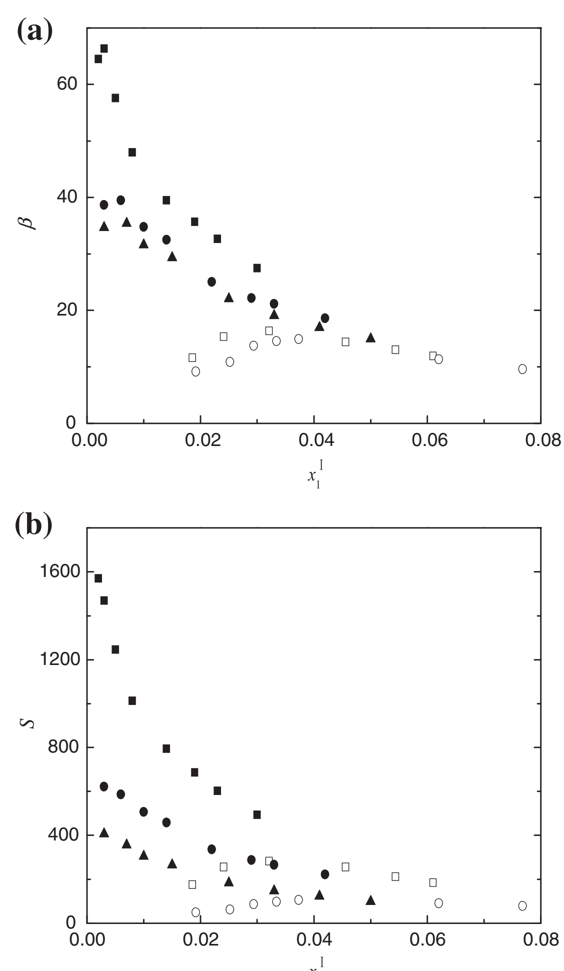

FIGURE 5. (a) Solute distribution ratios of methanol, β, and (b) methanol/hexane selectivities, S, as a function of the mole fraction of methanol in the upper layer, x_1^I, for the ternary systems {methanol (1) + hexane (2) + IL (3)} at T = 298.2 K and atmospheric pressure. ILs: ■, [MMIM][DMP]; ●, [EMIM][DEP]; ▲, [BMIM][DBP]; □, [BTMA][NTf$_2$] from reference [15]; ○, [TBMA][NTf$_2$] from reference [15].

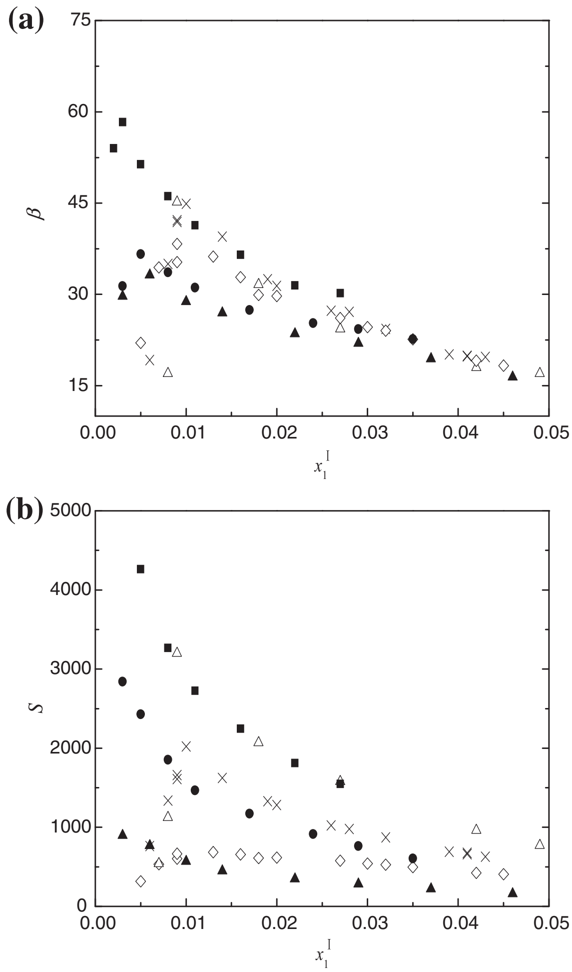

FIGURE 6. (a) Solute distribution ratios of methanol, β, and (b) methanol/heptane selectivities, S, as a function of the mole fraction of methanol in the upper layer, x_1^I, for the ternary systems {methanol (1) + heptane (2) + IL (3)} at T = 298.2 K and atmospheric pressure. ILs: ■, [MMIM][DMP]; ●, [EMIM][DEP]; ▲, [BMIM][DBP]; △, [COC_2mMOR][NTf_2] from reference [16]; ×, [COC_2mMOR][FAP] from reference [17]; ◇, [COC_2mPIP][FAP] from reference [17].

systems with hexane. The values of S for studied ILs are greatly higher than the unity, which indicates that the studied ILs can be considered as suitable solvents for the extraction of methanol from hexane and heptane.

On the other hand, comparisons with literature data for other ILs [15–17] are also included in figure 5 for the systems {methanol (1) + hexane (2) + IL (3)} and figure 6 for the systems {methanol (1) + heptane (2) + IL (3)}. As shown in figure 5, comparing the ternary systems {methanol (1) + hexane (2) + IL (3)}, the values of β for the ILs decrease in the order [MMIM][DMP] > [EMIM][DEP] > [BMIM][DBP] > [BTMA][NTf_2]> [TBMA][NTf_2], which means that smaller quantity of the studied ILs would be required for the extraction separation of methanol and hexane. And the values of S are greatly higher than the unity in all cases but decrease in the order [MMIM][DMP] > [EMIM][DEP] > [BTMA][NTf_2] > [BMIM][DBP] > [TBMA][NTf_2]. Besides, similar conclusions are derived from figure 6, where the comparison with the literature data of β and S of imide and morpholinium-based ILs is made for the ternary systems {methanol (1) + heptane (2) + IL (3)}. In figure 6, it is interesting to notice that the values of β for the ILs decrease in the following order [MMIM][DMP] > [COC_2mMOR][FAP] > [COC_2mPIP][FAP] > [COC_2mMOR][NTf_2] > [EMIM][DEP] > [BMIM][DBP], and the values of S decrease in the order [MMIM][DMP] > [COC_2mMOR][NTf_2] > [COC_2mMOR][FAP] > [EMIM][DEP] > [COC_2mPIP][FAP] > [BMIM][DBP].

The capabilities of the ILs to extract methanol from its mixtures with hexane and heptane are likely related to the Kamlet–Taft solvent parameters, which include the hydrogen-bond acidity, the hydrogen-bond basicity, and the polarity/polarizability [34]. The hydrogen-bond acidity value and hydrogen-bond basicity value, which express the ability to donate and accept hydrogen bonds, respectively, rely on the IL composition. The cation of an IL is responsible for the hydrogen-bond acidity value, and the hydrogen-bond basicity value is dependent on the anion of IL. For the systems methanol and alkanes, the extraction capabilities of ILs are probably mainly rely on the hydrogen-bond basicity [35]. The studied ILs (1.000) [36] with dialkylphosphate anions have higher hydrogen-bond basicity values than the ILs (0.243) [37] with imide-based anions (*i.e.*, [NTf_2]$^-$). On the other hand, the alkyl chain length of the ILs plays a negative role in the extraction capabilities of the ILs [13,14]. Thus, it can be understood that the [MMIM][DMP] obtains the highest values of β and S.

4. Conclusions

In this work, the performance of using three dialkylphosphate-based ILs as methanol extraction solvents was evaluated. For this purpose, the experimental LLE data for the ternary systems, {methanol (1) + hexane (2) + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP]) (3)}, and {methanol (1) + heptane (2) + ([MMIM][DMP], or [EMIM][DEP], or [BMIM][DBP]) (3)}, were measured at T = 298.2 K and atmospheric pressure. The experimental results were successfully correlated with the NRTL model. From the experimental LLE data, the methanol solute distribution ratios, β, and methanol/alkane selectivities, S, were calculated and compared with other values reported in the literature. On the whole, the values of β and S for the studied ILs decrease with the concentrations of methanol in the upper layer increase. The values of β and S are greatly higher than unity, especially at low mole fractions of methanol. As a result, the studied ILs can be used act as suitable solvents for the extraction of methanol from hexane and heptane, and being obtained a more efficient extraction process when heptane is involved. On the other hand, the investigated systems, especially for the ILs [MMIM][DMP], show higher the values of β, which indicate that a low amount of the IL would be required for the effective separation. Besides, the values of S decrease as the alkyl chain length of the ILs increases, and the values of S for the studied ILs decrease in the following order [MMIM][DMP] > [EMIM][DEP] > [BMIM][DBP], indicating that shorter chained ILs have a greater extracting potential. In conclusion, the IL [MMIM][DMP] is the best candidate for the separation of methanol from its mixtures with hexane and heptane in the extraction process.

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References

[1] A. Pucci, Pure Appl. Chem. 61 (1989) 1363–1372.
[2] L. Laroche, H.W. Andersen, M. Morari, N. Bekiaris, Can. J. Chem. Eng. 69 (1991) 1302–1319.
[3] A. Nann, C. Held, G. Sadowski, Ind. Eng. Chem. Res. 52 (2013) 18472–18481.
[4] J.M. Lopes, A.V.M. Nunes, M.N. da Ponte, Z.P. Visak, V. Najdanovic-Visak, Ind. Eng. Chem. Res. 53 (2014) 9883–9888.
[5] M.S. Al-Tuwaim, K.H.A.E. Alkhaldi, M.S. Fandary, A.S. Al-Jimaz, J. Chem. Thermodyn. 43 (2011) 1804–1809.

[6] C.M.S.S. Neves, J.F.O. Granjo, M.G. Freire, A. Robertson, N.M.C. Oliveira, J.A.P. Coutinho, Green Chem. 13 (2011) 1517–1526.
[7] M.T.G. Jongmans, B. Schuur, A.B. de Haan, Ind. Eng. Chem. Res. 50 (2011) 10800–10810.
[8] J.G. Huddleston, A.E. Visser, W.M. Reichert, H.D. Willauer, G.A. Brokera, R.D. Rogers, Green Chem. 3 (2001) 156–164.
[9] L.P.N. Rebelo, J.N.C. Lopes, J.M.S.S. Esperanca, H.J.R. Guedes, J. Lachwa, V. Najdanovic-Visak, Z.P. Visak, Acc. Chem. Res. 40 (2007) 1114–1121.
[10] A.B. Pereiro, A. Rodríguez, Ind. Eng. Chem. Res. 48 (2009) 1579–1585.
[11] A.B. Pereiro, F.J. Deive, J.M.S.S. Esperanca, A. Rodríguez, Fluid Phase Equilib. 294 (2010) 49–53.
[12] A.L. Revelli, F. Mutelet, J.N. Jaubert, J. Chem. Eng. Data 56 (2011) 3873–3880.
[13] B. González, S. Corderí, A.G. Santamaría, J. Chem. Thermodyn. 60 (2013) 9–14.
[14] B. González, S. Corderí, Fluid Phase Equilib. 354 (2013) 89–94.
[15] N.M. Aranda, B. González, J. Chem. Thermodyn. 68 (2014) 32–39.
[16] A. Marciniak, M. Królikowski, Fluid Phase Equilib. 318 (2012) 56–60.
[17] M. Wlazło, A. Marciniak, Fluid Phase Equilib. 338 (2013) 253–256.
[18] X. Chen, F. Cai, X. Wu, C. Asumana, G. Yu, J. Chem. Eng. Data 58 (2013) 1186–1192.
[19] F. Cai, X. Wu, C. Chen, X. Chen, C. Asumana, M.R. Haque, G. Yu, Fluid Phase Equilib. 352 (2013) 47–53.
[20] F. Cai, J.J. Ibrahim, L. Gao, R. Wei, G. Xiao, Fluid Phase Equilib. 382 (2014) 254–259.
[21] F. Cai, J.J. Ibrahim, L. Niu, W. Xu, G. Xiao, J. Chem. Eng. Data (2014), http://dx.doi.org/10.1021/je500672d.
[22] F. Cai, G. Xiao, Fluid Phase Equilib. 386 (2015) 155–161.
[23] F. Cai, M. Zhao, Y. Wang, F. Wang, G. Xiao, J. Chem. Thermodyn. 81 (2015) 177–183.
[24] H. Renon, J.M. Prausnitz, AIChE J. 14 (1968) 135–144.
[25] Y.H. Zhou, A.J. Robertson, J.H. Hillhouse, Phosphonium and imidazolium salts and methods of their preparation, WO2004/016631, 2004-02-26.
[26] A.V. Orchillés, P.J. Miguel, E. Vercher, A. Martínez-Andreu, J. Chem. Eng. Data 55 (2010) 1669–1674.
[27] A.V. Orchillés, P.J. Miguel, V. González-Alfaro, E. Vercher, A. Martínez-Andreu, J. Chem. Eng. Data 56 (2011) 4454–4460.
[28] A.V. Orchillés, P.J. Miguel, V. González-Alfaro, E. Vercher, A. Martínez-Andreu, J. Chem. Eng. Data 57 (2012) 394–399.
[29] S. Lago, H. Rodríguez, A. Soto, A. Arce, J. Chem. Eng. Data 56 (2011) 1273–1281.
[30] M. Larriba, P. Navarro, J. García, F. Rodríguez, Ind. Eng. Chem. Res. 52 (2013) 2714–2720.
[31] G.W. Meindersma, A.J.G. Podt, A.B. de Haan, Fluid Phase Equilib. 247 (2006) 158–168.
[32] M. Larriba, P. Navarro, J. García, F. Rodríguez, Fluid Phase Equilib. 364 (2014) 48–54.
[33] E. Vercher, F.J. Rojo, A. Martínez-Andreu, J. Chem. Eng. Data 44 (1999) 1216–1221.
[34] A. Brandt, J.P. Hallett, D.J. Leak, R.J. Murphy, T. Welton, Green Chem. 12 (2010) 672–679.
[35] J. Palgunadi, S.Y. Hong, J.K. Lee, H. Lee, S.D. Lee, M. Cheong, H.S. Kim, J. Phys. Chem. B 115 (2011) 1067–1074.
[36] Y. Fukaya, K. Hayashi, M. Wada, H. Ohno, Green Chem. 10 (2008) 44–46.
[37] L. Crowhurst, P.R. Mawdsley, J.M. Perez-Arlandis, P.A. Salter, T. Welton, Phys. Chem. Chem. Phys. 5 (2003) 2790–2794.

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